The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Section-Centered Hidden-Response
Observer-Relevant Pullback-Operator Core
Collapse to an Observer-Relevant Pullback Body-Operator Core

Abstract

The current active-primary theorem already removed the full observer-relevant pullback-quadratic core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb}}$ as the primitive stopping point of the active section-centered hidden-response line by collapsing its benchmark-facing content to the smaller observer-relevant pullback-operator core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}$. The preferred next move below that frontier would be to derive or justify that operator core from still more primitive explicit substrate structure in a way that genuinely changes benchmark-facing status. The present note takes the honest fallback route available before such a cleaner deeper law is fixed: it proves a strictly smaller theorem-level collapse inside the current operator core itself.

The positive result is that, relative to the fixed proto-section benchmark base, the current operator core still contains presentational redundancy. The compact section-centered displacement body already recovers the ambient displacement space canonically as the fiberwise span of its displacement fibers, so the separate ambient-space listing does not add independent benchmark-facing freedom. What survives is the smaller observer-relevant pullback body-operator core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}$: the displacement body together with the smooth positive self-adjoint pullback operator family and benchmark faithfulness.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. The positive result is therefore a genuine smaller theorem-level collapse strictly inside the current operator-core frontier. What remains open is to derive or justify the pullback body-operator core from still more primitive explicit substrate structure, or else to prove a still sharper collapse, necessity result, or uniqueness result strictly inside that smaller core.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity again reaches a point where depth alone is not the honest default move. The current active-primary theorem already showed that the full observer-relevant pullback-quadratic core is not the live primitive stopping point at current scope, because the quadratic pullback functional and the exact-shell presentation are recovered from the smaller observer-relevant pullback-operator core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}$. That theorem therefore spent the pullback-quadratic-core presentation as the live burden.

The present note asks the next narrower question available strictly inside that operator core. Once the fixed proto-section benchmark base is held fixed, does the current operator core still contain an independently live ambient-space presentation, or is that ambient space already forced

by the displacement body on which the current operator acts? The present note proves that the ambient-space presentation is redundant. Because each displacement fiber has nonempty interior in the listed ambient space, the body already spans that space. The live benchmark-facing datum therefore contracts one level further to the pullback body-operator core.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not introduce a second operative metric, it does not enlarge the low-energy matter law, and it does not reopen the larger pullback-quadratic-core, hidden-response, residual-normal-form, or pre-proto-section presentations as if they were again independently live. Its narrower benchmark-facing role is to identify whether the separate ambient displacement-space listing inside the current operator core is still independently meaningful at benchmark scope.

2. Fixed Inputs

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Definition 2 (Recorded proto-section benchmark base and downstream chain)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The active line already fixes the following proto-section benchmark base and downstream consequences.*

1. *the current proto-section benchmark base* $\mathfrak{Q}^{\mathrm{sub,proto,secgen}}$*, presented through*

$$\mathcal{Q}_\sigma^{\mathrm{psec}}, \qquad \mathcal{B}_\sigma^{\mathrm{psec}}, \qquad \mathscr{E}_\sigma^{\mathrm{psec}}, \qquad \mathcal{Z}_\sigma^{\mathrm{psec}};$$

2. *the current observer-relevant pullback-operator core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}$*, the current observer-relevant pullback-quadratic core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb}}$*, the current section-centered residual-normal-form realization* $\mathfrak{R}^{\mathrm{sub,proto,presec,res}}$*, the current observer-relevant pre-proto-section minimizer-section core* $\mathfrak{R}^{\mathrm{sub,proto,presec,sec}}$*, and the current benchmark-faithful pre-proto-section package* $\mathfrak{R}^{\mathrm{sub,proto,presec}}$ *as already recorded on the active line;*

3. *the previously recorded fact that, once a benchmark-faithful observer-relevant pullback-operator core over the fixed proto-section benchmark base is available, the current pullback-quadratic-core realization, the current section-centered residual-normal-form realization, the current pre-proto-section package, the current minimizer-section core, and the previously recorded seed, root, foundation, ground, origin, precursor, lift, shell-restriction, split-core, selector-law, combined-response, ambient-package, trace-core, exact-shell-affine-nucleus, continuity-Hessian compressed core, and benchmark one-metric observer-package consequences follow unchanged;*

4. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

3. Current Operator-Core Frontier

Definition 3 (Observer-relevant pullback-operator core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* benchmark-faithful observer-relevant pullback-operator core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}$ *for sector* σ *consists of:*

1. *a finite-dimensional Euclidean section-centered displacement space*

$$V_\sigma^{\mathrm{pppresec}};$$

2. *a compact convex section-centered displacement body*

$$\mathcal{D}_\sigma^{\mathrm{pppresec}} \subset \mathcal{B}_\sigma^{\mathrm{psec}} \times V_\sigma^{\mathrm{pppresec}}$$

with nonempty interior in each fiber

$$\mathcal{D}_\sigma^{\mathrm{pppresec}}(y) \equiv \{v \in V_\sigma^{\mathrm{pppresec}} : (y, v) \in \mathcal{D}_\sigma^{\mathrm{pppresec}}\},$$

where every fiber is nonempty, compact, and convex, and

$$(y, 0) \in \mathcal{D}_\sigma^{\mathrm{pppresec}} \qquad \textit{for every } y \in \mathcal{B}_\sigma^{\mathrm{psec}};$$

3. *a smooth family of positive self-adjoint pullback operators*

$$A_\sigma^{\mathrm{pppresec,pb}}(y) : V_\sigma^{\mathrm{pppresec}} \to V_\sigma^{\mathrm{pppresec}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$

satisfying

$$\left\langle v, A_\sigma^{\mathrm{pppresec,pb}}(y) v \right\rangle > 0 \qquad \textit{for every } y \in \mathcal{B}_\sigma^{\mathrm{psec}}, \quad 0 \neq v \in V_\sigma^{\mathrm{pppresec}};$$

4. *benchmark faithfulness in the sense of Definition 2(4).*

From this operator core define the recovered pullback quadratic data by

$$\mathscr{P}_\sigma^{\mathrm{pppresec,pb}}(y,v) \equiv \frac{1}{2}\left\langle v, A_\sigma^{\mathrm{pppresec,pb}}(y)v\right\rangle, \qquad \widehat{\mathcal{Z}}_\sigma^{\mathrm{pppresec,pb,bodyop}} \equiv \{(y,0)\in \mathcal{D}_\sigma^{\mathrm{pppresec}} : y \in \mathcal{Z}_\sigma^{\mathrm{psec}}\}.$$

Remark 1. *Definition 3 is the full current operator-core presentation. The present note asks whether the separate ambient displacement-space listing remains independently live once the displacement body and pullback operator family are already fixed.*

4. Observer-Relevant Pullback Body-Operator Core

Definition 4 (Observer-relevant pullback body-operator core)**.** *Fix $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ and a benchmark-faithful pullback-operator core in the sense of Definition 3. The* observer-relevant pullback body-operator core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}$ *for sector σ consists only of:*

1. *the compact convex section-centered displacement body*

$$\mathcal{D}_\sigma^{\mathrm{pppresec}} \subset \mathcal{B}_\sigma^{\mathrm{psec}} \times V_\sigma^{\mathrm{pppresec}};$$

2. *the smooth family of positive self-adjoint pullback operators*

$$A_\sigma^{\mathrm{pppresec,pb}}(y) : V_\sigma^{\mathrm{pppresec}} \to V_\sigma^{\mathrm{pppresec}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}};$$

3. *benchmark faithfulness in the sense of Definition 2(4).*

From this smaller core define the recovered ambient displacement space by

$$\widehat{V}_\sigma^{\mathrm{pppresec,pb,bodyop}} \equiv \mathrm{span}\left(\bigcup_{y\in\mathcal{B}_\sigma^{\mathrm{psec}}} \mathcal{D}_\sigma^{\mathrm{pppresec}}(y)\right) \subset V_\sigma^{\mathrm{pppresec}}.$$

The ambient-space listing is therefore treated as derived rather than separately primitive.

Remark 2. *Definition 4 is strictly smaller than Definition 3 at current scope. Once the displacement body and the pullback operator family are fixed, the only remaining question is whether the ambient displacement space adds any further benchmark-facing freedom. The present theorem shows that it does not.*

5. Collapse of the Operator Core

Proposition 1 (The displacement body recovers the ambient displacement space)**.** *For every benchmark-faithful pullback-operator core and every $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$,*

$$\widehat{V}_\sigma^{\mathrm{pppresec,pb,bodyop}} = V_\sigma^{\mathrm{pppresec}}.$$

Proof. Fix $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$. By Definition 3(2), the fiber $\mathcal{D}_\sigma^{\mathrm{pppresec}}(y)$ has nonempty interior in the ambient finite-dimensional Euclidean space $V_\sigma^{\mathrm{pppresec}}$. Any subset of a finite-dimensional vector space with nonempty interior spans that vector space. Hence

$$\mathrm{span}\big(\mathcal{D}_\sigma^{\mathrm{pppresec}}(y)\big) = V_\sigma^{\mathrm{pppresec}}.$$

Because that fiber is contained in the union appearing in Definition 4,

$$V_\sigma^{\mathrm{pppresec}} \subset \widehat{V}_\sigma^{\mathrm{pppresec,pb,bodyop}} \subset V_\sigma^{\mathrm{pppresec}},$$

which proves the claim. □

Proposition 2 (The full operator core is reconstructed from the body-operator core)**.** *For every benchmark-faithful pullback-operator core, the full observer-relevant pullback-operator core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}$ *is reconstructed from the smaller pullback body-operator core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}$ *together with the fixed proto-section benchmark base.*

Proof. Proposition 1 recovers the ambient displacement space exactly from the displacement body. The body itself and the pullback operator family are already part of Definition 4, and benchmark faithfulness is retained there as well. Therefore every constituent of Definition 3 is recovered from the smaller body-operator core plus the fixed proto-section benchmark base. □

Proposition 3 (The pullback quadratic data remain canonical)**.** *For every benchmark-faithful pullback-operator core and every* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, the pullback quadratic data are recovered canonically from the body-operator core by*

$$\mathscr{P}_\sigma^{\mathrm{pppresec,pb}}(y,v) \equiv \frac{1}{2}\left\langle v, A_\sigma^{\mathrm{pppresec,pb}}(y)v\right\rangle, \qquad \widehat{\mathcal{Z}}_\sigma^{\mathrm{pppresec,pb,bodyop}} \equiv \{(y,0) \in \mathcal{D}_\sigma^{\mathrm{pppresec}} : y \in \mathcal{Z}_\sigma^{\mathrm{psec}}\}.$$

These agree with the current pullback quadratic data of Definition 3.

Proof. The formula for the recovered quadratic functional uses only the operator family retained in Definition 4. The shell formula uses only the retained displacement body together with the fixed proto-section shell. Since the body-operator core reconstructs the full operator core by Proposition 2, these recovered pullback quadratic data are exactly the same as those already carried by Definition 3. □

Proposition 4 (All current downstream uses factor through the body-operator core)**.** *For every benchmark-faithful pullback-operator core, all current benchmark-facing downstream uses of the current operator-core frontier already factor through the smaller pullback body-operator core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}$ *together with the fixed proto-section benchmark base. In particular, the current pullback-quadratic core, the current section-centered residual-normal-form realization, the current pre-proto-section package, the current minimizer-section core, and the previously recorded seed, root, foundation, ground, origin, precursor, lift, shell-restriction, split-core, selector-law, combined-response, ambient-package, trace-core, exact-shell-affine-nucleus, continuity-Hessian compressed core, and benchmark one-metric observer-package consequences follow unchanged once the body-operator core is fixed.*

Proof. By Proposition 2, the full operator core is recovered from the smaller body-operator core. Proposition 3 then recovers the same pullback quadratic data. Definition 2(3) records that, once a benchmark-faithful operator core over the fixed proto-section benchmark base is available, all of the listed downstream consequences follow unchanged. Therefore every current benchmark-facing downstream use of the operator-core frontier factors through the smaller body-operator core together with the fixed proto-section benchmark base. □

Proposition 5 (Separate ambient-space presentation is benchmark neutral at current scope)**.** *For every benchmark-faithful pullback-operator core, the separate presentation of the ambient displacement space*

$$V_\sigma^{\mathrm{pppresec}}$$

does not add independent benchmark-facing freedom beyond the smaller observer-relevant pullback body-operator core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}$.

Proof. By Proposition 1, the ambient displacement space is reconstructed canonically from the displacement body. By Proposition 4, all current benchmark-facing downstream uses of the operator-core frontier already factor through the smaller body-operator core once that reconstruction is made. Therefore the separate ambient-space listing does not provide additional benchmark-facing freedom beyond $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}$. □

Proposition 6 (The benchmark package remains fixed)**.** *Every observer-relevant pullback body-operator core preserves the same benchmark one-metric package* (1)*. In particular, the operative metric remains exactly* $g^\sharp$*, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is part of benchmark faithfulness in Definition 4(3). The present note only removes presentational redundancy inside the current pullback-operator-core frontier. It does not alter the benchmark exact-closure package. □

6. Main Theorem

Theorem 1 (Observer-relevant pullback-operator core collapse to an observer-relevant pullback body-operator core)**.** *Fix the primitive continuation data of Definition 1, the recorded proto-section benchmark base and downstream chain of Definition 2, and a benchmark-faithful pullback-operator core in the sense of Definition 3. Then, for each sector* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, the live benchmark-facing burden inside the current pullback-operator-core frontier collapses from the observer-relevant pullback-operator core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}$ *to the smaller observer-relevant pullback body-operator core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}$*. More precisely:*

1. *the ambient displacement space is recovered canonically from the fiberwise span of the displacement body by Proposition 1;*
2. *the full operator-core presentation is therefore reconstructed from the body-operator core and the fixed proto-section benchmark base by Proposition 2;*
3. *the same pullback quadratic data remain canonical by Proposition 3;*
4. *all current benchmark-facing downstream uses of the operator-core frontier factor through the body-operator core by Proposition 4;*
5. *the separate ambient-space presentation adds no independent benchmark-facing freedom beyond the body-operator core by Proposition 5;*
6. *the benchmark boundary remains fixed by Proposition 6;*

7. *consequently the remaining honest burden below the previous pullback-operator-core frontier is no longer the full observer-relevant pullback-operator core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}$ *as such. What remains is to derive or justify the sharper pullback body-operator core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}$ *from still more primitive substrate structure, or else to prove a still smaller collapse, sharper necessity result, or sharper uniqueness result strictly inside that smaller core.*

Proof. Item (1) is Proposition 1. Item (2) is Proposition 2. Item (3) is Proposition 3. Item (4) is Proposition 4. Item (5) is Proposition 5. Item (6) is Proposition 6. Item (7) is the routing consequence of the previous items together with the active safeguard against counting another same-output deeper-origin relay as new front-line progress when the live benchmark-relevant datum is unchanged. □

Corollary 1 (What must be done next)**.** *After Theorem 1, the next honest benchmark-facing manuscript must either:*

1. *derive or justify the observer-relevant pullback body-operator core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}$ *from still more primitive explicit substrate structure in a way that changes benchmark-facing status;*

2. *identify a genuinely smaller theorem-level observer-relevant datum strictly inside* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}$ *and prove a sharper collapse, sharper necessity result, or sharper uniqueness result there;*

3. *do either of the previous two while preserving the same benchmark one-metric package, the same ordinary-matter route, and the same claim boundary.*

Another same-output deeper-origin relay that preserves this body-operator core and therefore merely reproduces the same current pullback-operator core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}$*, the same current pullback-quadratic core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb}}$*, the same current section-centered hidden-response package, the same current residual-normal-form datum, the same current benchmark-faithful pre-proto-section package, and the same downstream benchmark-facing consequences, another re-expansion back to the larger pullback-operator-core, pullback-quadratic-core, hidden-response, residual-normal-form, or pre-proto-section presentations as if their recovered constituents were still independently live, or any move that introduces a second operative metric, enlarges the ordinary matter law, or uses stronger promotion language does not satisfy the present burden.*

Proof. Theorem 1 shows that the ambient-space presentation is no longer part of the minimal benchmark-facing datum at this stage. Therefore a subsequent manuscript must improve on the smaller body-operator core rather than merely restating the larger operator core. □

7. Conclusion

The positive result of this note is a disciplined second compression step inside the current pullback frontier. The previous active-primary theorem already showed that the full pullback-quadratic-core presentation is not the minimal benchmark-facing datum, because its benchmark-relevant content collapses to an observer-relevant pullback-operator core. The present theorem then takes the honest next internal route available there: it shows that even that operator-core presentation is still larger than necessary.

What survives as the live burden is the observer-relevant pullback body-operator core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}$. Relative to the already fixed proto-section benchmark base, that sharper datum is just the compact

section-centered displacement body together with the smooth positive self-adjoint pullback operator family acting on it. The ambient displacement space is already recovered as the span of the displacement body, so its separate listing does not add independent benchmark-facing freedom.

This is the honest next burden after the present theorem. A new primary manuscript must derive or justify that sharper body-operator core from still more primitive explicit substrate structure, or else collapse it further, while preserving the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary. The current result does not yet complete a first-principles derivation of GR from substrate microphysics, but it does narrow the live derivational burden one level further on the active line.